explore_relations_20261101

November 1, 2026

0.1 Using relational model

[1]:
```
import re

import polars as pl

from tool_extraction import SEED, artifact_siblings

artifact_siblings_with_content = (
    artifact_siblings
    .select("entry_sha", "content", "repo_full_name", "artifact_path",␣
 ↪"entry_name", "content_fetched")
    .filter(pl.col("content_fetched") == 1)
    .with_columns(pl.col("content").str.strip_chars().replace("", None))
    .filter(pl.col("content").is_not_null())
    .drop("content_fetched")
)

# Every column gets the same seed so the sampled rows stay aligned.
sample = artifact_siblings_with_content.select(pl.all().sample(n=500,␣
 ↪seed=SEED)).collect()

MAX_LENGTH = 384
MAX_WORDS_CONTEXT = 5

def get_snippet(text: str, entity: str) -> list[str]:
    pattern = re.compile(rf"(?<!\w){re.escape(entity)}(?!\w)")
    snippets = []
    for match in pattern.finditer(text):
        before = text[: match.start()].split()
        after = text[match.end() :].split()
        highlighted = f"[{match.group()}]"
        snippet = " ".join(before[-MAX_WORDS_CONTEXT:] + [highlighted] + after[:
 ↪MAX_WORDS_CONTEXT])
        if len(before) > MAX_WORDS_CONTEXT:
            snippet = "..." + snippet
        if len(after) > MAX_WORDS_CONTEXT:
            snippet = snippet + "..."
```

```
        snippets.append(snippet)
    return snippets
```

[2]:
```
from gliner import GLiNER
import polars as pl
import seaborn_polars as snl

relational_model = GLiNER.from_pretrained("knowledgator/gliner-relex-large-v0.
 ↪5")

HIT_THRESHOLD = 0.6
HIT_RELATION_THRESHOLD = 0.6
HIT_LABELS = ["tool", "organization"]
HIT_RELATION_LABELS = ["from_organization"]
HIT_DTYPES = pl.Struct({
    "snippet": pl.List(pl.String),
    "entity": pl.String,
    "label": pl.String,
    "relation": pl.String,
    "related": pl.String,
    "related_label": pl.String,
    "score": pl.Float64,
})

def extract(text: str) -> list[dict]:
    clipped = (text or "")[:MAX_LENGTH]
    entities, relations = relational_model.inference(
        [clipped],
        labels=HIT_LABELS,
        relations=HIT_RELATION_LABELS,
        threshold=HIT_THRESHOLD,
        relation_threshold=HIT_RELATION_THRESHOLD,
    )
    return [
        {
            "entity": rel["head"]["text"],
            "label": rel["head"]["type"],
            "relation": rel["relation"],
            "related": rel["tail"]["text"],
            "related_label": rel["tail"]["type"],
            "score": rel["score"],
            "snippet": get_snippet(clipped, rel["head"]["text"]),
        }
        for rel in relations[0]
    ]

hits = (
```

```
    sample.lazy()
    .select("entry_sha", "content")
    .with_columns(
        pl.col("content")
        .map_elements(extract, return_dtype=pl.List(HIT_DTYPES))
        .alias("hits")
    )
    .explode("hits")
    .unnest("hits")
    .with_columns(
        pl.col("entity").str.to_lowercase(),
        pl.col("related").str.to_lowercase(),
        pl.when(pl.col("snippet").list.len() == 0)
        .then(pl.lit(["(no match)"]))
        .otherwise(pl.col("snippet"))
        .alias("snippet"),
    )
    .select(
        "entry_sha", "snippet", "entity", "label",
        "relation", "related", "related_label", "score",
    )
)

edges = (
    hits.group_by("entity", "label", "relation", "related", "related_label")
    .agg(
        pl.col("entry_sha").n_unique().alias("skills"),
        pl.col("score").mean().alias("score"),
    )
)

nodes = (
    pl.concat([
        hits.select(pl.col("entity").alias("name"), pl.col("label")),
        hits.select(pl.col("related").alias("name"), pl.col("related_label").
 ↪alias("label")),
    ])
    .group_by("name")
    .agg(pl.col("label").mode().first().alias("label"))
)

review = hits.explode("snippet").head(20).collect()
with pl.Config(fmt_str_lengths=200, tbl_rows=20):
    display(review)
```

```
/home/nichmorgan/Projects/nau/msr-data-
challenge-2027/scripts/.venv/lib/python3.13/site-
packages/torch/jit/_script.py:1491: FutureWarning: `torch.jit.script` is
```

```
deprecated. Please switch to `torch.compile` or `torch.export`.
  warnings.warn(
/home/nichmorgan/Projects/nau/msr-data-
challenge-2027/scripts/.venv/lib/python3.13/site-
packages/huggingface_hub/utils/_validators.py:190: UserWarning: The
`resume_download` argument is deprecated and ignored in `snapshot_download`.
Downloads always resume whenever possible.
  warnings.warn(
Downloading bytes:           |  0.00B
Reconstructing (incomplete total…): |          |  0.00B /  0.00B
Fetching 12 files:   0%|          | 0/12 [00:00<?, ?it/s]
shape: (20, 8)
 entry_sha    snippet      entity       label        relation    related    ␣
 ↪related_l   score
 ---          ---          ---          ---          ---         ---         abel ␣
 ↪      ---
 str          str          str          str          str         str         --- ␣
 ↪      f64
                                                                            str  ␣
 ↪
 69301b4ae0   #!/usr/bin   python3      tool         from_orga   word        tool ␣
 ↪      0.828872
 10905ede5a   /env                                  nization                     ␣
 ↪
 063048ca53   [python3]                                                         ␣
 ↪
 ebcc7caabf   """Apply                                                         ␣
 ↪
              lifecycle                                                        ␣
 ↪
              edits to                                                         ␣
 ↪
              existing..                                                       ␣
 ↪
              .                                                                ␣
 ↪
 2b1ef3ef6a   #!/usr/bin   python3      tool         from_orga   dataclass  ␣
 ↪organizat   0.887571
 d9d6faa346   /env                                  nization    e           ion  ␣
 ↪
 bc02026a33   [python3]                                                        ␣
 ↪
```

```
b6a592dd24   """                                                                ␣
↪
             Validate                                                          ␣
↪
             MudBlazor                                                         ␣
↪
             CSS variab                                                        ␣
↪
             les…                                                              ␣
↪
2e09f6a20c   …tunnels   tunnel      tool        from_orga   cloudflar ␣
↪organizat   0.995089
c44dfa35c8   .read desc                          nization    e          ion   ␣
↪
198f04d653   ription:                                                          ␣
↪
f47fac397b   Get a                                                             ␣
↪
             Cloudflare                                                        ␣
↪
             [Tunnel]                                                          ␣
↪
             by ID perm                                                        ␣
↪
             ission:                                                           ␣
↪
             tunnel.rea                                                        ␣
↪
             d request:                                                        ␣
↪
             …                                                                 ␣
↪
2e09f6a20c   …request   tunnel_id   tool        from_orga   cloudflar ␣
↪organizat   0.994947
c44dfa35c8   : method:                          nization    e          ion   ␣
↪
198f04d653   GET path:                                                         ␣
↪
f47fac397b   /accounts/                                                        ␣
↪
             {account_i                                                        ␣
↪
             d}/cfd_tun                                                        ␣
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             nel/{ [tun                                                        ␣
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            nel_id] }                                                  ␣
↪
            inputs: -                                                  ␣
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            name: acco                                                 ␣
↪
            unt_id…                                                    ␣
↪
2e09f6a20c  …true     tunnel_id   tool        from_orga   cloudflar ␣
↪organizat  0.994947
c44dfa35c8  location:                           nization   e          ion ␣
↪
198f04d653  path -                                                     ␣
↪
f47fac397b  name: [tun                                                 ␣
↪
            nel_id]                                                    ␣
↪
            type:                                                      ␣
↪
            string                                                     ␣
↪
            required:                                                  ␣
↪
            true locat                                                 ␣
↪
            ion:…                                                      ␣
↪
e674f7086f  #!/bin/bas  meeting-pr  organizati  from_orga  euo       ␣
↪organizat  0.601902
a7426e60d8  h # [meeti  ep          on          nization              ion ␣
↪
d3823c74ce  ng-prep]                                                   ␣
↪
6cfa83ccde  /scripts/p                                                 ␣
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            rep.sh -                                                   ␣
↪
            Research                                                   ␣
↪
            attendees                                                  ␣
↪
            for…                                                       ␣
↪
e674f7086f  …CONFIG   meeting-pr  organizati  from_orga  euo       ␣
↪organizat  0.601902
```

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a7426e60d8  & ARGS ---  ep          on          nization              ion
d3823c74ce  PREP_DIR="
6cfa83ccde  ${PREP_DIR
            :-$HOME/.c
            onfig/ [me
            eting-prep
            ] }" CONFI
            G_FILE="$P
            REP_DIR/co
            nfig.json"
            HISTORY_FI
            LE="$PREP_
            DIR/brief-
            history.js
            on" LOG_FI
            LE="$PREP_
            DIR/prep-l
            og.json"
            BRIEFS_DI…
e674f7086f  …specifi  pipefail    tool        from_orga  meeting-p
organizat   0.874535
a7426e60d8  c meeting                        nization   rep        ion
d3823c74ce  (on-demand
6cfa83ccde  ) set -euo
```

```
            [pipefail]                                                  ␣
↪
            # ---                                                       ␣
↪
            CONFIG &                                                    ␣
↪
            ARGS…                                                       ␣
↪
9f0d13f8e9  # [Chain    chain       tool        from_orga   cha         ␣
↪organizat  0.95773
da916bdf7a  Analysis]   analysis                nization                ion ␣
↪
896447656d  Use this                                                     ␣
↪
08d58293e1  when the                                                    ␣
↪
            user…                                                       ␣
↪
17cb6801c9  {           skills.git  tool        from_orga   github      ␣
↪organizat  0.910305
fd232fedd4  "source":                           nization                ion ␣
↪
e553b5d593  "https://g                                                  ␣
↪
7ff39f25b5  ithub.com/                                                  ␣
↪
            mattpocock                                                  ␣
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            / [skills.                                                  ␣
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            git] ",                                                     ␣
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            "sourceTyp                                                  ␣
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            e": "git",                                                  ␣
↪
            "repoUrl":                                                  ␣
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            "https://g                                                  ␣
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            ithub.com/                                                  ␣
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            mattpocock                                                  ␣
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            /skills.gi                                                  ␣
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            t",…                                                                ␣
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17cb6801c9  …"https:  skills.git  tool        from_orga  github    ␣
↪organizat  0.910305
fd232fedd4  //github.c                        nization              ion  ␣
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e553b5d593  om/mattpoc                                                          ␣
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7ff39f25b5  ock/skills                                                          ␣
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            .git", "so                                                          ␣
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            urceType":                                                          ␣
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            "git",                                                              ␣
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            "repoUrl":                                                          ␣
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            "https://g                                                          ␣
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            ithub.com/                                                          ␣
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            mattpocock                                                          ␣
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            / [skills.                                                          ␣
↪
            git] ",                                                             ␣
↪
            "subpath":                                                          ␣
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            "migrate-t                                                          ␣
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            o-shoehorn                                                          ␣
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            ", "instal                                                          ␣
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            ledAt":                                                             ␣
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            "2026-04-1                                                          ␣
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            4T10:06:51                                                          ␣
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            .370…                                                               ␣
↪
17cb6801c9  {           skills.git  tool        from_orga  mattpococ  ␣
↪organizat  0.974121
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fd232fedd4  "source":                          nization   k          ion
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e553b5d593  "https://g
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7ff39f25b5  ithub.com/
↪
            mattpocock
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            e": "git",
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            "repoUrl":
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            "https://g
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            ithub.com/
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            t",…
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17cb6801c9  …"https:  skills.git  tool       from_orga  mattpococ
↪organizat  0.974121
fd232fedd4  //github.c                       nization   k          ion
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e553b5d593  om/mattpoc
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7ff39f25b5  ock/skills
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            urceType":
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            "git",
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            "repoUrl":
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            "https://g
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              ithub.com/                                                  ␣
↪
              mattpocock                                                  ␣
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              / [skills.                                                  ␣
↪
              git] ",                                                     ␣
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              "subpath":                                                  ␣
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              "migrate-t                                                  ␣
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              o-shoehorn                                                  ␣
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              ", "instal                                                  ␣
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              ledAt":                                                     ␣
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              "2026-04-1                                                  ␣
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              4T10:06:51                                                  ␣
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              .370…                                                       ␣
↪
17cb6801c9    …"git",    migrate-to    tool          from_orga    github     ␣
↪organizat    0.920482
fd232fedd4    "repoUrl":    -shoehorn                  nization                 ion  ␣
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e553b5d593    "https://g                                                    ␣
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7ff39f25b5    ithub.com/                                                  ␣
↪
              mattpocock                                                  ␣
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              /skills.gi                                                  ␣
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              t",                                                         ␣
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              "subpath":                                                  ␣
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            stalledAt"                                                  ␣
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            : "2026-04                                                  ␣
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            -14T10:06:                                                  ␣
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            51.370Z" }                                                  ␣
↪
17cb6801c9  …"git",   migrate-to  tool        from_orga  mattpococ  ␣
↪organizat  0.969477
fd232fedd4  "repoUrl":  -shoehorn               nization   k          ion  ␣
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e553b5d593  "https://g                                                  ␣
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            mattpocock                                                  ␣
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            /skills.gi                                                  ␣
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            t",                                                         ␣
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            "subpath":                                                  ␣
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            " [migrate                                                  ␣
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            rn] ", "in                                                  ␣
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            stalledAt"                                                  ␣
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            : "2026-04                                                  ␣
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            -14T10:06:                                                  ␣
↪
            51.370Z" }                                                  ␣
↪
1fa3a63ba7  interface:  gstack-rev  tool        from_orga  gstack-re  ␣
↪organizat  0.628223
3ac8ebd384  display_na  iew                     nization   view       ion  ␣
↪
3f14f512a2  me: " [gst                                                  ␣
↪
7957710a09  ack-review                                                  ␣
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            ] " short_                                                  ␣
↪
            descriptio                                                  ␣
↪
            n: "" defa                                                  ␣
↪
            ult_prompt                                                  ␣
↪
            : "Use…                                                     ␣
↪
1fa3a63ba7  …"gstack  gstack-rev  tool        from_orga  gstack-re  ␣
↪organizat  0.628223
3ac8ebd384  -review"    iew                   nization   view       ion  ␣
↪
3f14f512a2  short_desc                                                  ␣
↪
7957710a09  ription:                                                    ␣
↪
            "" default                                                  ␣
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            _prompt:                                                    ␣
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            "Use [gsta                                                  ␣
↪
            ck-review]                                                  ␣
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            for this                                                    ␣
↪
            task."                                                      ␣
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            policy:                                                     ␣
↪
            allow_impl                                                  ␣
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            icit_invoc                                                  ␣
↪
            ation:…                                                     ␣
↪
2ceff63c21  # [Claude   claude      tool        from_orga  ssot      ␣
↪organizat  0.889648
8590189637  Session     session               nization              ion  ␣
↪
8137c14d28  Exporter]   exporter                                        ␣
↪
7dfac1e9d3  -                                                           ␣
↪
```

```
            Companion                                                  ␣
↪
            Bridge ##                                                  ␣
↪
            Purpose…                                                   ␣
↪
2ceff63c21  # [Claude   claude      tool        from_orga  skill.md   tool ␣
↪     0.866045
8590189637  Session     session                 nization               ␣
↪
8137c14d28  Exporter]   exporter                                       ␣
↪
7dfac1e9d3  -                                                          ␣
↪
            Companion                                                  ␣
↪
            Bridge ##                                                  ␣
↪
            Purpose…                                                   ␣
↪
eeacdd7dd2  …with: #  ggplot2     tool        from_orga  r           ␣
↪organizat  0.972707
8ef8a3bf35  R 4.3.0 #                          nization              ion  ␣
↪
49890a7cb8  [ggplot2]                                                 ␣
↪
cc0703a0ae  3.4.2 #                                                    ␣
↪
            tidytext                                                   ␣
↪
            0.4.1 #…                                                   ␣
↪
```